Analysis: Analyze trends, correlations and anomalies

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Executive summary

• revenue and cost show a strong positive correlation (r=1.00, n=80, p=1.3e-89).

• quantity and unit_price show a strong negative correlation (r=-0.91, n=80, p=5e-32).

• unit_price and cost show a strong positive correlation (r=0.90, n=80, p=3.4e-29).

• unit_price has 16 anomalous value(s) (20.0% of rows) by robust z-score; most extreme: 50,000 at row 0.

• quantity has 14 anomalous value(s) (17.5% of rows) by robust z-score; most extreme: 100 at row 55.

Rows	Columns	Quality score	Evidence verified
80	11	99.9	8/8

Key findings (evidence-verified)

✔ revenue and cost show a strong positive correlation (r=1.00, n=80, p=1.3e-89).

✔ quantity and unit_price show a strong negative correlation (r=-0.91, n=80, p=5e-32).

✔ unit_price and cost show a strong positive correlation (r=0.90, n=80, p=3.4e-29).

✔ unit_price has 16 anomalous value(s) (20.0% of rows) by robust z-score; most extreme: 50,000 at row 0.

✔ quantity has 14 anomalous value(s) (17.5% of rows) by robust z-score; most extreme: 100 at row 55.

✔ unit_price differs significantly across product (ANOVA, p=0): highest in 'Laptop' (mean 50,000) vs lowest in 'Notebook' (mean 100.0).

✔ quantity differs significantly across category (ANOVA, p=1.3e-26): highest in 'Office' (mean 60.77) vs lowest in 'Electronics' (mean 4.42).

✔ 'Laptop' accounts for 48.4% of total revenue by product.

Recommendations

→ Fix data quality: 5 extreme outliers (3×IQR) in 'quantity' (suggested action: clip outliers).

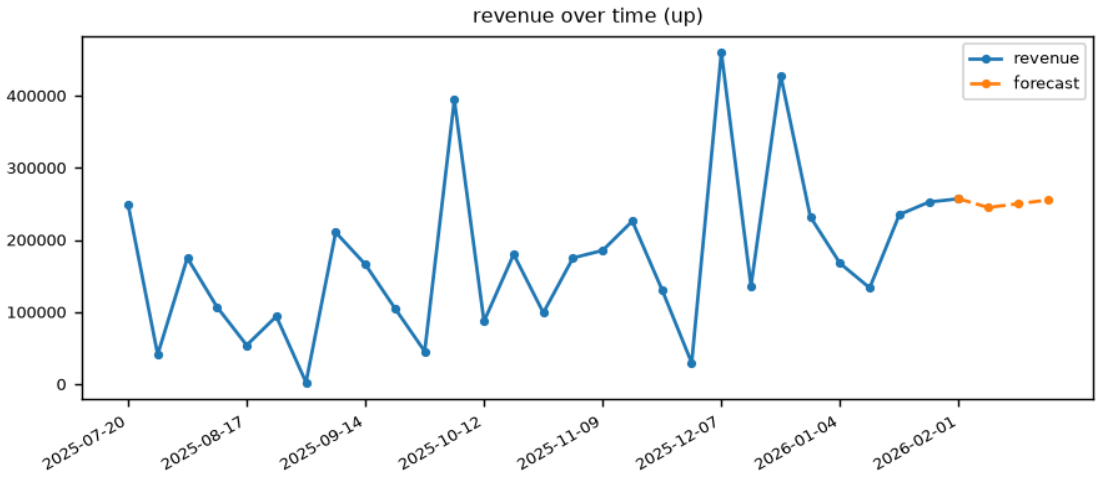
→ Fix data quality: 1 extreme outliers (3×IQR) in 'cost' (suggested action: clip outliers).

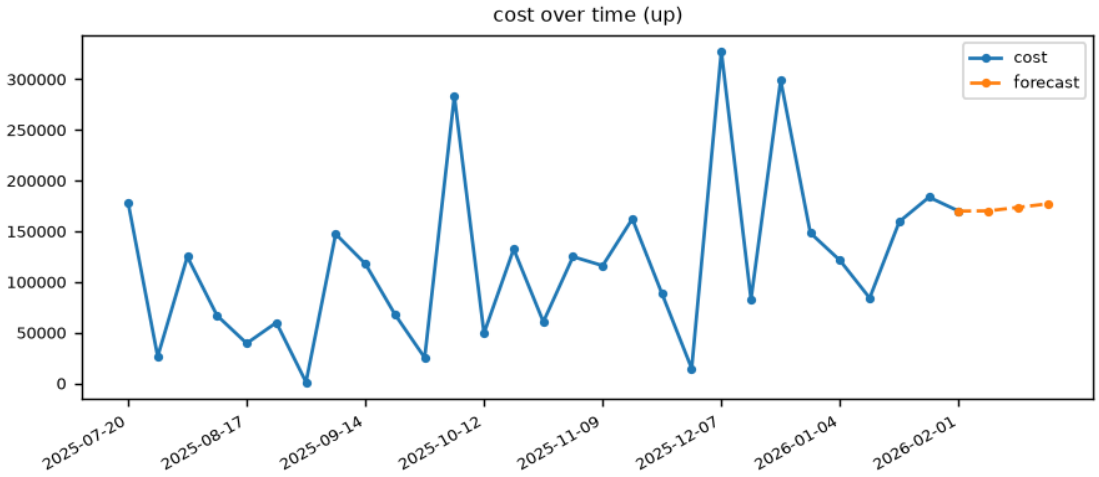
→ Test whether the strongly correlated variables are causally related (e.g. controlled experiment) before acting.

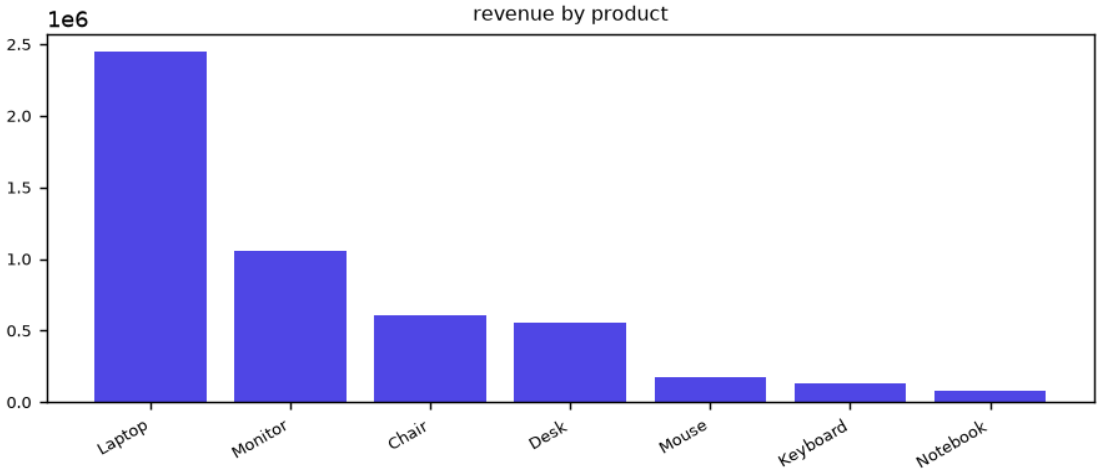
→ Review anomalous records for data-entry errors or genuine exceptional events before relying on averages.

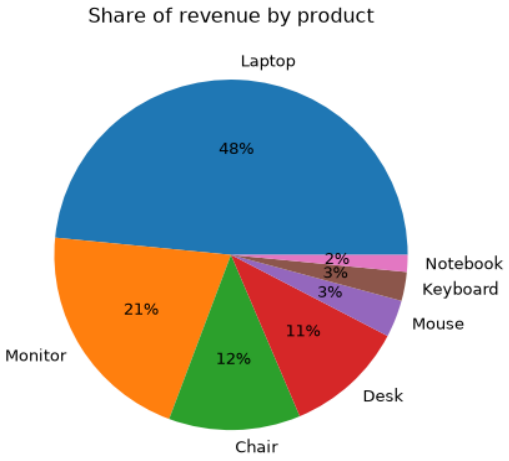
→ Replicate best-performing group practices in lower-performing groups.

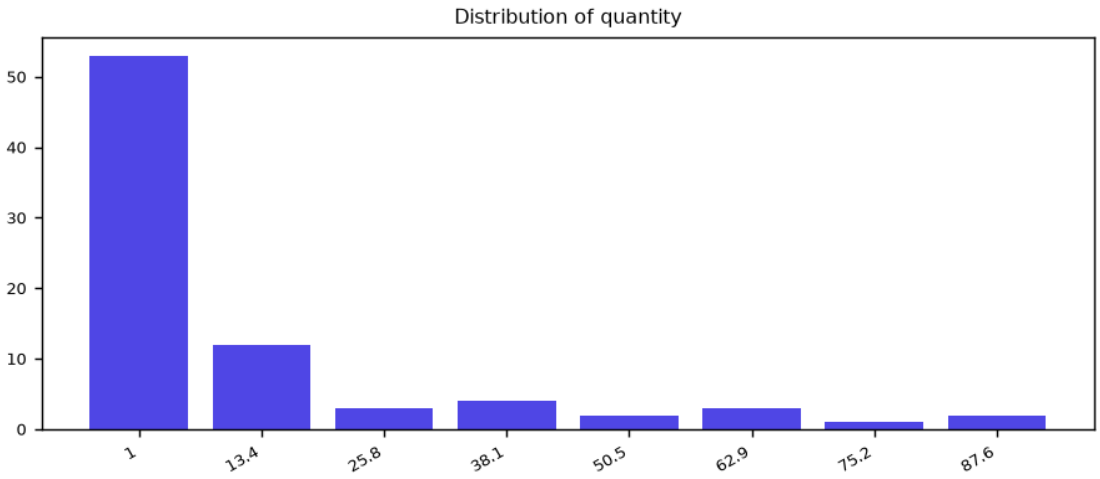
→ High concentration detected: assess dependency risk and diversification options.


revenue over time (up)
revenue
forecast
400000
300000
200000
100000
0
2025-07-20
2025-08-17
2025-09-14
2025-10-12
2025-11-09
2025-12-07
2026-01-04
2026-02-01


cost over time (up)
cost
forecast
300000
250000
200000
150000
100000
50000
0
2025-07-20
2025-08-17
2025-09-14
2025-10-12
2025-11-09
2025-12-07
2026-01-04
2026-02-01


revenue by product
1e6
2.5
2.0
1.5
1.0
0.5
0.0
Laptop
Monitor
Chair
Desk
Mouse
Keyboard
Notebook





Validation

8 of 8 numeric claims independently re-computed and confirmed.